

Dr. Lenard Dome

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RESEARCH

I am a computational cognitive scientist at the University of Tübingen. My research develops and evaluates process models of learning, generalisation, and decision-making – with a central commitment to model adequacy; *whether models capture the right computational mechanisms, not merely the right outcomes*.

I am increasingly interested in what formal models of cognition reveal about the conditions under which decision-making succeeds or fails, both as a diagnostic lens for *psychiatric conditions* and as a principled basis for designing systems that support rather than supplant human judgment.

ACADEMIC POSITIONS

2023 - Postdoctoral researcher. University of Tuebingen. (Advisor: Tobias Hauser)
2023 - Guest Scientist. Max Planck Institute for Biological Cybernetics.
2020 - 2023 Ph.D. Psychology. Plymouth University. (Supervisor: Andy Wills).
2021 - 2023 Visiting Lecturer, Tutor and Specialist. University of Plymouth.
2019 - 2020 M.Sc. Advanced Psychology. Plymouth University. (Supervisor: Patric Bach).
2019 - 2023 Teaching Assistant. University of Plymouth.
2017 - 2019 Research Assistant. (Line Manager: Andy Wills).
2017 - 2018 Research Assistant. (Line Manager: Marina Wimmer).

EDUCATION

2020 - 2023 Ph.D. Psychology. Computational Cognitive Science. *Straight Pass*.
2019 - 2020 M.Sc. Advanced Psychology. Plymouth University. *Distinction*.
2017 - 2019 B.Sc. Psychology. Plymouth University. *First*.

THESIS

Dome, L. (2023) *Testing the adequacy of formal models of an irrational learning effect*. University of Plymouth, UK.
Defended: 2023.09.12. <https://pearl.plymouth.ac.uk/psy-theses/74>

GRANTS AND AWARDS

- 2026 **Awarded** DZPG Travel Grant.
- 2020 **Awarded** [Undergraduate Project Prize](#). Experimental Psychological Society / British Science Association.
- 2019 **Awarded** Doctoral Training Studentship (1+3). ESRC & SWDTP. £87,510.62.

PUBLICATIONS

- in.prep. **Dome, L.** “Shared attention vectors are unsustainable under multi-outcome learning.”
- 2026 **Dome, L.,** Wills, A.J. “Errorless Irrationality: A unified computational account of the inverse base-rate effect across predictive, observational, and unsupervised procedures.” *arXiv*. [10.48550/arXiv.2608.06149](#)
- 2026 **Dome, L.,** Moses-Payne, M. E., Skvortsova, V., & Hauser, T. U., “Age-related shifts in metacognition reveal converging confidence for correct and error judgments across the lifespan.” *PsyArXiv*. [10.31234/osf.io/ugb9s](#)
- 2026 **Dome, L.,** Kadri, K., Wagner, B. J., Webb, A. & Hauser, T. U. “cpm: a python library for theory-driven modeling in Computational Psychiatry.” *PLOS Computational Biology*. [10.1371/journal.pcbi.1014481](#).
- 2026 Schlegelmilch, R. & **Dome, L.** Snake oil or Panacea? How to Misuse AI in Scientific Inquiries of the Human Mind. *Behavioural Sciences*. [10.3390/bs16020219](#).
- 2026 Marzuki, A., Kosina, L., **Dome, L.,** Hewitt, S., & Hauser, T. Metacognitive antecedents to states of mental ill-health: Drops in confidence precede symptoms of OCD. *Nature Communications*. (in press). [10.21203/rs.3.rs-7544256/v1](#).
- 2025 **Dome, L.** & Wills, A.J. “g-Distance: On the comparison of model and human heterogeneity.” *Psychological Review*. [10.1037/rev0000550](#)
- 2025 **Dome, L.** & Wills, A.J. “Better generalization through distraction? Concurrent load reduces the size of the inverse base-rate effect.” *Psychonomic Bulletin & Review*. [10.3758/s13423-025-02661-1](#)
- 2023 **Dome, L.** & Wills, A. J. “Errorless irrationality: removing error-driven components from the inverse base-rate effect paradigm.” *Proceedings of the 45th Annual Conference of the Cognitive Science Society*. Austin, TX: Cognitive Science Society. [escholarship.org/uc/item/0kw671vv](#)
- 2021 **Dome, L.,** Edmunds, C. E. R., & Wills, A. J. “SUSTAIN captures category learning, recognition, and hippocampal activation in a unidimensional vs information-integration task.” *Proceedings of the 43rd Annual Conference of the Cognitive Science Society*. Austin, TX: Cognitive Science Society. [escholarship.org/uc/item/5r98q3dr](#)
- Spicer, S. G., Wills, A. J., Jones, P. M., Mitchell, C., & **Dome, L.** “Representing uncertainty in the Rescorla-Wagner model: blocking, the redundancy effect, and outcome base rate.” *Open Journal of Experimental Psychology and Neuroscience*. [doi.org/10.46221/ojepn.2021.6623](#)
- 2019 Wimmer, M. C., **Dome, L.,** Wennekers, T. & Hancock, P. J. B. “Is the letter cancellation task a suitable index of ego-depletion? Empirical and conceptual issues.” *Social Psychology*. [doi.org/10.1027/1864-9335/a000393](#)

RESEARCH SOFTWARE

-  **psp** *maintainer, developer* Implements an n -dimensional parameter space partitioning algorithm for evaluating the global behaviour of formal computational models as described by Pitt, Kim, Navarro and Myung (2006).

18K downloads. [\[source code\]](#) [\[cran\]](#) doi: [10.32614/CRAN.package.psp](https://doi.org/10.32614/CRAN.package.psp)

 **cpm** *maintainer, developer* cpm is a fundamental python package for Computational Psychiatry. It is designed to provide a set of tools for researchers and clinicians to analyze and model data in the field of Computational Psychiatry. [\[source code\]](#)

 **catlearn** *senior developer* Catlearn is an archive of formal models of categorization and learning, plus benchmark datasets to test them against. It currently houses 14 formal mathematical models of learning and categorization. I have contributed 8 models to the package. 57K downloads. [\[source code\]](#) [\[cran\]](#) doi: [10.32614/CRAN.package.catlearn](https://doi.org/10.32614/CRAN.package.catlearn)

INVITED TALKS

2026 **Yale University (USA)**

MAP Talk. *cpm*: a Python library for theory-driven modeling in computational psychiatry.

2024 **University of Tübingen (Germany)**

Psychiatry Colloquium. *cpm*: a Python library for theory-driven modeling in computational psychiatry.

2021 **British Science Association / Experimental Psychological Society (UK)**

Undergraduate Project Prize Talk (2020). Clearing confounds from the inverse base-rate effect: irrationality and concurrent load. [Video](#).

Liverpool John Moores University (UK)

Cognitive and Affective Neuroscience Group Seminar. The role of base rates in categorizing ambiguity and forming stereotypes.

CONFERENCE PRESENTATION AND TALKS

2026 Research Colloquium. Tuebingen, Germany.

2026 German Mental Health Centre (DZPG) ECS Congress. Berlin, Germany.

2025 Computational Psychiatry Conference. Tuebingen, Germany.

2023 45th Annual Conference of the Cognitive Science Society. Sydney, Australia.

2023 MathPsych/ICCM/EMPG: Amsterdam, The Netherlands.

2023 Meeting of the Experimental Psychology Society. Plymouth.

2022 22nd Associative Learning Symposium: Gregynog Hall, Wales, UK.

2021 Cognitive Science Conference: Online.

2019 The British Psychological Society: South West Undergraduate Conference. Plymouth, UK.

SUPERVISION AND TEACHING

2025–2026 **Max Planck Institute for Biological Cybernetics (Germany)** [CaCTüS](#) internship student supervision.

2024–2026 **University of Tübingen (Germany)** Seminar: Introduction to computational modeling (B.Sc., M.Sc.).

2025 **University of Tübingen (Germany)** Workshop: Introduction to computational modeling (M.Sc., Ph.D., Postdocs).

2022–2023 **University of Plymouth (UK)** Workshop: Introduction to quantitative analysis with R (DClinPsy).

2021–2023 **University of Plymouth (UK)** Final-year project supervision (B.Sc.).

2021–2023 **University of Plymouth (UK)** Seminar: Bayesian sample size estimation // Data visualization in R (B.Sc.).

2021–2023 **University of Plymouth (UK)** Research methods in R (B.Sc.).

SOCIETY MEMBERSHIPS

Cognitive Science Society Member
Experimental Psychological Society
German Centrum for Mental Health (DZPG) ECS
European Brain and Behaviour Society

PEER-REVIEW

The American Statistician.